Nikolaos Vardalakis, PhD

Neuroscientist | Human Intracranial Electrophysiology & Neural Biomarkers
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PROFILE

Neuroscientist and engineer developing electrophysiological biomarkers from human intracranial recordings (stereo-EEG, ECoG, LFP) in clinical neuromodulation studies. I own analyses end to end, from raw recordings and signal-quality QC to rigorous statistics for small, repeated-measures clinical datasets, and work closely with neurosurgeons and clinical teams. Background in electrical engineering, computational modeling and primate electrophysiology.

EXPERIENCE

Postdoctoral Researcher, Department of Neurosurgery, University of Pennsylvania, Philadelphia, PA Jul 2024 – Present

- Lead biomarker analyses of intracranial and scalp EEG for **DBS and TMS studies** in treatment-refractory OCD, substance use disorder and loss-of-control eating, in close collaboration with neurosurgeons and clinical teams.
- Developed **connectivity and spectral biomarkers that predict DBS response in OCD**; built the decoding and statistics pipeline (regularized logistic regression, grouped cross-validation, permutation testing, FDR control).
- Designed analysis plans and **mixed-effects / cluster-based permutation** frameworks for small-sample intracranial data, covering evoked connectivity (CCEPs), phase synchrony and time–frequency effects.
- Built reproducible preprocessing and **signal-quality QC pipelines**: re-referencing, artifact removal, and automated bad-channel and bad-trial rejection.

Doctoral Researcher, Institute of Neurodegenerative Diseases (CNRS), University of Bordeaux, France Sep 2020 – Apr 2024

- Designed and ran **non-human primate ECoG/SEEG experiments** testing how hippocampal stimulation affects memory; co-developed a multi-camera neuronavigation system for deep-brain electrode implantation.
- Built a spiking-network model predicting how stimulation alters **theta–gamma phase–amplitude coupling** (first author, *eLife* 2024).

EDUCATION

Ph.D., Neuroscience, University of Bordeaux, France 2024
M.Sc., Biomedical Engineering, Delft University of Technology, the Netherlands 2020
M.Eng./B.Sc., Electrical and Computer Engineering (Signal Processing), Technical University of Crete, Greece 2017

SKILLS

Machine learning: Feature engineering from neural time series, regularized classifiers, grouped cross-validation, AUROC
Statistics: Mixed-effects models, small-sample and repeated-measures inference, experimental design
Signal processing: Spectral analysis, coherence and phase synchrony, phase–amplitude coupling, evoked potentials
Recordings: Human stereo-EEG, ECoG, LFP, scalp EEG; microelectrode recordings; non-human primate electrophysiology
Programming: Python (pandas, scikit-learn, NumPy, SciPy), MATLAB (FieldTrip), C/C++, Git, AI-assisted development
Clinical: Deep brain stimulation, TMS, closed-loop neuromodulation, human-subjects research in psychiatry

SELECTED PUBLICATIONS

- **Vardalakis N.**, Aussel A., Rougier N.P., Wagner F.B. A dynamical computational model of theta generation in hippocampal circuits to study theta–gamma oscillations during neurostimulation. *eLife* (2024).
- Gupta A., Boissenin A., **Vardalakis N.**, et al. High-accuracy electrode implantation in deep brain structures using multi-camera neuronavigation in non-human primates. *J. Neural Eng.* (2025).
- Gupta A., **Vardalakis N.**, Wagner F.B. Neuroprosthetics: from sensorimotor to cognitive disorders. *Communications Biology* (2023).
- Seilheimer R.L., **Vardalakis N.**, et al. Patient-specific network connectivity predicts deep brain stimulation response in obsessive-compulsive disorder. *In preparation.* (+2 more in preparation)